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Experimental and detailed kinetic modeling study of 1-pentanol oxidation in a JSR and combustion in a bomb

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Abstract

1-Pentanol is among the possible alcohols usable as an alternative to conventional gasoline and diesel fuels. In order to better understand the combustion characteristics of 1-pentanol, this study presents new experimental data for 1-pentanol in two complementary experimental configurations. Stable species concentration profiles were measured in a jet-stirred reactor (JSR) at 10 atm over a range of equivalence ratios and temperatures. Laminar flame speeds of 1-pentanol-air premixed laminar flame were measured at 1 atm and 423 K for equivalence ratios of 0.7–1.4. The oxidation of 1-pentanol in these experimental configurations was modeled using an extended detailed chemical kinetic reaction mechanism (2099 reactions involving 261 species) derived from a previously proposed scheme for the oxidation of methanol-, ethanol-, and 1-butanol-based fuels. The proposed mechanism shows good agreement with the present experimental data. Reaction path analyses were conducted to interpret the JSR results.
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Keywords: 1-Pentanol; Alcohol; Flame speed; Jet-stirred reactor; Kinetic modeling

1. Introduction

Because they are renewable, biofuels are attracting great interest as transportation fuels. They can be locally produced, may be less polluting, sometimes more biodegradable, and could reduce net greenhouse gas emissions [1]. Ethanol accounts for over 90% of all biofuels' production worldwide [2]. However, mixing stability issues may appear with simple alcohols whereas larger alcohols would mix better with petrol-derived fuels thanks to their longer alkyl carbon chain. Since 1-butanol was announced to be sold soon as a gasoline blending constituent [3], Dagaut and Togbé studied the oxidation of butanol-gasoline surrogate mixtures (85–15 vol%) in a JSR at 10 atm and a kinetic reaction mechanism was derived for modeling the oxidation of butanol-gasoline surrogate mixtures [4]. 1-Pentanol is among the longer carbon-chain alcohols that could be blended with conventional fuels. However, so far, it received little attention since only engine experiments were reported in the literature [5,6] whereas bio-pentanol could be produced [7,8].

The goal of this study is to provide new experimental data on 1-pentanol oxidation in

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complementary well-defined laboratory experiments. Stable species concentration profiles were obtained for 1-pentanol oxidation in JSR at 10 atm ($10.13.10^5$ Pa) over a range of equivalence ratios (0.35–4.0) and temperatures. Also, burning velocities were measured for 1-pentanol premixed laminar flames at various equivalence ratios and 1 atm. This new data set is used to validate an extended detailed chemical kinetic model of the oxidation of C_1–C_5 alcohols-based fuels.

2. Experimental

2.1. Jet-stirred reactor

The jet-stirred reactor (JSR) experimental setup used here has been described earlier [4,9]. The reactor consists of a 4 cm diameter fused silica sphere equipped with four nozzles of 1 mm i.d.. Prior to the injectors, the reactants were diluted with nitrogen (<100 ppm H_2O, <50 ppm O_2, <1000 ppm Ar, <5 ppm H_2) and mixed. A high degree of dilution (0.1–0.15 mol.% of fuel) was used, reducing temperature gradients and heat release in the JSR. The reactants were high-purity oxygen (99.995% pure) and high-purity 1-pentanol (>99% pure from Aldrich, CAS 71-41-0), which was sonically degassed before use. The reactants were preheated before injection to minimize temperature gradients inside the reactor. A Shimadzu LC10 AD VP pump with an online degasser (Shimadzu DGU-20 A3) was used to deliver the fuel to an atomizer–vaporizer assembly maintained at 200 °C. Good thermal homogeneity along the vertical axis of the reactor (gradients of ca. 1 K/cm) was observed during the experiments by thermocouple measurements (0.1 mm Pt–Pt/Rh-10%, located inside a thin-wall silica tube). The reacting mixtures were sampled by a movable fused silica low pressure sonic probe, and then sent to the analyzers via a Teflon heated line (200 °C). They were analyzed online by FTIR (10 m path length, 200 mBar, resolution of 0.5 cm^{-1}) and off-line after collection and storage in 1 L Pyrex bulbs. Off-line analyses were done using gas chromatographs (GC) equipped with capillary columns (DB-624, CP-Al2O3-KCl, and Carboplot-P7), a TCD (thermal conductivity detector), and an FID (flame ionization detector). A GC-MS (Varian quadrupole V1200) operating with electron ionization (70 eV) was used for products identification.

The experiments were performed at steady state, at a constant mean residence time, τ, of 0.7 s and a constant pressure of 10 atm, the reactants continually flowing in the reactor, whereas the temperature of the gases inside the JSR was increased stepwise. A good repeatability of the measurements and a reasonably good carbon balance (typically $100 \pm 10\%$) were obtained in this series of experiments.

2.2. Laminar flame speed setup

The device to measure the laminar flame speed consists of a stainless steel cylindrical combustion chamber with an inside volume of 24.32 L. Four windows provide optical accesses into the chamber. Before filling, a vacuum is created inside the chamber. The volume of air is introduced into the chamber with a thermal mass flow meter. High-purity 1-pentanol (>99% pure), which is liquid at ambient pressure and temperature, is injected though a solenoid injector. The exact volume to be injected is measured precisely ($\pm$0.005 g). The injection conditions of the solenoid injector (frequency and gate) are adjusted to finish the injection process before the filling process of the air. Before entering the chamber, the 1-pentanol/air mixture is heated up till 500 K to ensure a perfect vaporization of the liquid. The temperature of the chamber has to be elevated enough to avoid any condensation of the fuel on the walls. This temperature was chosen close to the 1-pentanol's boiling temperature (410.95 K): $T_{ini} = 423$ K. In these conditions the quantity of fuel injected is fully vaporized for all the equivalence ratios studied in the present work. An electric fan, located inside the chamber, mixes all the gases. A delay before ignition avoids any perturbation during the flame propagation. Two tungsten electrodes linked to a conventional capacitive discharge ignition system, are used to form the spark gap (2.8 mm) at the center of this chamber. Measurements are limited to flames having diameters less than 60 mm, implying that the total volume of burnt gases is less than 0.5% of the chamber volume. In this initial part of the flame expansion, the total chamber pressure can be considered constant. Observation times are less than 25 ms for all the cases investigated (depending on the equivalence ratio). A range of equivalence ratios, from 0.7 to 1.4, is tested. Each case of equivalence ratio is repeated five times. In the following, the mean value is presented. The standard deviation, corresponding to the scattering in experiments, is also indicated.

The image of the flame is obtained by the classical shadowgraphy technique. The parallel light from an Ar-Ion laser source is created by two plano-convex lenses (25 and 1000 mm focal lengths, respectively). The shadowgraphic images are recorded by using a high speed video CMOS camera (Photron APX) operating at 6000 frames per second with an exposure time of 20 μs. The temporal evolution of the expanding spherical flame is then analyzed. A global schematic view of the system was presented previously [10].

2.3. Laminar burning velocity extraction

The stretched spatial flame velocity is determined from the measured flame radius (R_f) against

time (t) by $s_b = \frac{dR_f}{dt}$. A propagating flame surface undergoes both curvature and flow field aerodynamics (strain) effects which lead to modifications of the flame surface. In the case of a spherical expanding laminar flame, the total stretch rate acting on the flame is defined as $\kappa = 2\frac{S_b}{R_f}$. By taking into account a relation linking the propagation speed and the stretch rate, the un-stretched propagation speed (s_b^0) can be determined. In the present study a non-linear relation is considered $\left(\frac{S_b}{S_b^0}\right) ln\left(\frac{S_b}{S_b^0}\right) = -\frac{2L_bK}{S_b^0}$ [11], where L_b is the burned gas Markstein length and S_b^0 is the un-stretched propagation speed. The use of this non-linear formulation affords better results especially for lean conditions (where the Markstein length is far from unity). Then, for a constant pressure flame propagation, the laminar burning velocity can be deduced from the un-stretched propagation speed and the density ratio by using the following equation: $S_u^0 = \frac{\rho_b}{\rho_u} S_b^0$, with ρ_b and ρ_b, respectively the density of burned and unburned gases. The densities were evaluated using the Cantera code [12].

3. Computational methods

The kinetic modeling for 1-pentanol oxidation in the two experimental setups was performed using the CHEMKIN package [13] and the Cantera computer code [12]. The JSR was modeled using the perfectly stirred reactor (PSR) code [14], and the laminar flame speeds were modeled using the premix flame code Cantera. The inputs to each simulation include a detailed chemical kinetic reaction mechanism, a dataset of thermochemical properties, and a dataset of transport properties. These input files are available as supplementary material to this publication and from the authors.

The kinetic reaction mechanism used here is based on a previously proposed mechanism for the oxidation methanol, ethanol, and 1-butanol

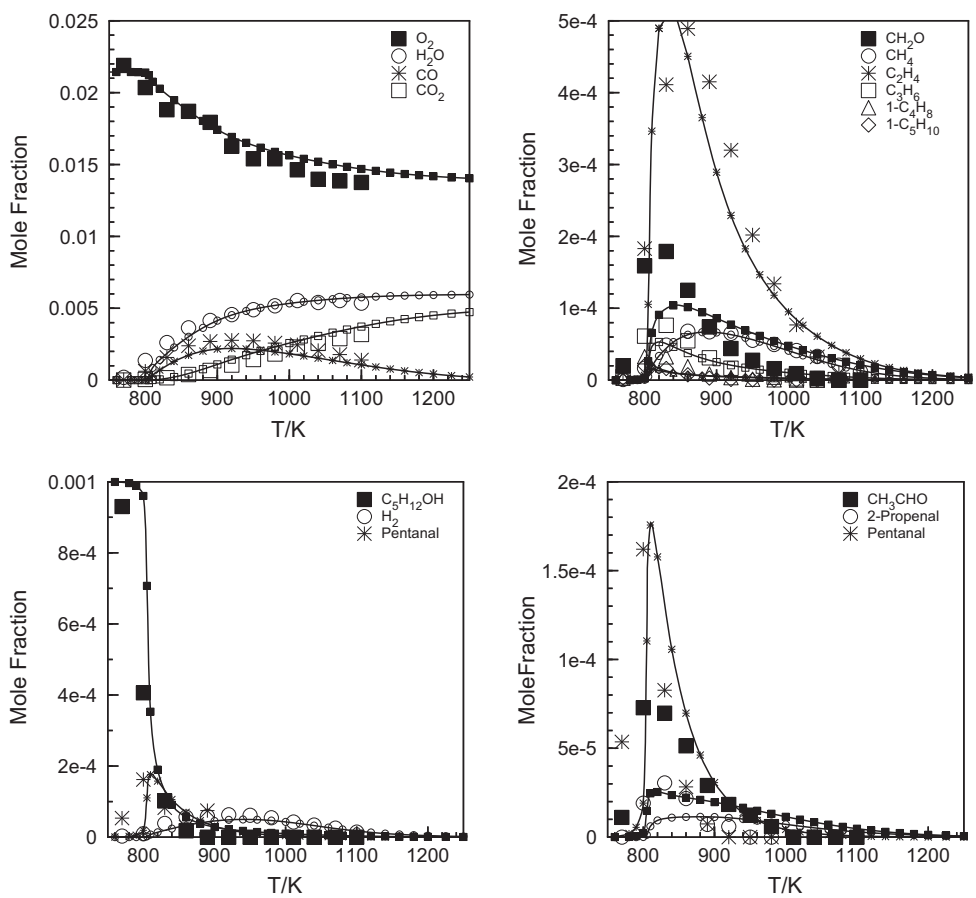

Fig. 1. Experimental (large symbols) and computed (lines and small symbols) concentration profiles obtained from the oxidation of 1-pentanol in a JSR at $\phi = 0.35$, $P = 10$ atm, $\tau = 0.7$ s.

[4,10,15,16] based fuels in JSR at 1–10 atm. It was extended by including the chemistry of 1-pentanol oxidation to that of C_1–C_4 alcohols. The proposed mechanism consists of 2099 reactions involving 261 species. For the new species the thermochemical properties were computed using group and bond additivity methods [17]. 1-Pentanol oxidation scheme is similar to that of 1-butanol [4,15]. It initially reacts via thermal decomposition involving C–O, C–C, and C–H bond breaking ($C_5H_{11}OH\text{-}1 \Leftrightarrow OH + aC_5H_{11}$, $C_5H_{11}OH\text{-}1 \Leftrightarrow H + C_5H_{11}O$, $C_5H_{11}OH\text{-}1 \Leftrightarrow 1C_4H_9 + CH_2OH$, $C_5H_{11}OH\text{-}1 \Leftrightarrow nC_3H_7 + C_2H_4OH$, $C_5H_{11}OH\text{-}1 \Leftrightarrow C_2H_5 + cC_3H_6OH$, $C_5H_{11}OH\text{-}1 \Leftrightarrow CH_3 + dC_4H_8OH$).

Propagation reactions occur via H-atom abstraction with simple species X (H, O, OH, HO_2, CH_3, HCO, CH_2O, CH_3O, C_2H_5, aC_3H_5, and C_4H_7) yielding hydroxypentyl and pentoxy radicals ($C_5H_{11}OH\text{-}1 + X \Leftrightarrow HX + C_5H_{11}O$, $C_5H_{11}OH\text{-}1 + X \Leftrightarrow HX + a\text{–}eC_5H_{10}OH$, where $aC_5H_{10}OH$ represents 1-hydroxy-1-pentyl and $eC_5H_{10}OH$ 1-hydroxy-5-pentyl). The isomerization reactions of these radicals were also included ($aC_5H_{10}OH \Leftrightarrow cC_5H_{10}OH$, $aC_5H_{10}OH \Leftrightarrow dC_5H_{10}OH$, $aC_5H_{10}OH \Leftrightarrow eC_5H_{10}OH$, $dC_5H_{10}OH \Leftrightarrow bC_5H_{10}OH$, $eC_5H_{10}OH \Leftrightarrow cC_5H_{10}OH$, $eC_5H_{10}OH \Leftrightarrow bC_5H_{10}OH$, $eC_5H_{10}OH \Leftrightarrow C_5H_{11}O$, $dC_5H_{10}OH \Leftrightarrow C_5H_{11}O$, $cC_5H_{10}OH \Leftrightarrow C_5H_{11}O$) as well as their decomposition reactions and reaction with O_2. The rate constants used here are consistent with those used previously for modeling the oxidation of 1-butanol oxidation in similar conditions [4,15].

4. Results and discussion

4.1 Jet-stirred reactor

The experiments were performed at a constant mean residence time of 0.7 s and a constant pressure of 10 atm. The equivalence ratio, ϕ, was varied from fuel lean to fuel rich conditions (0.35,

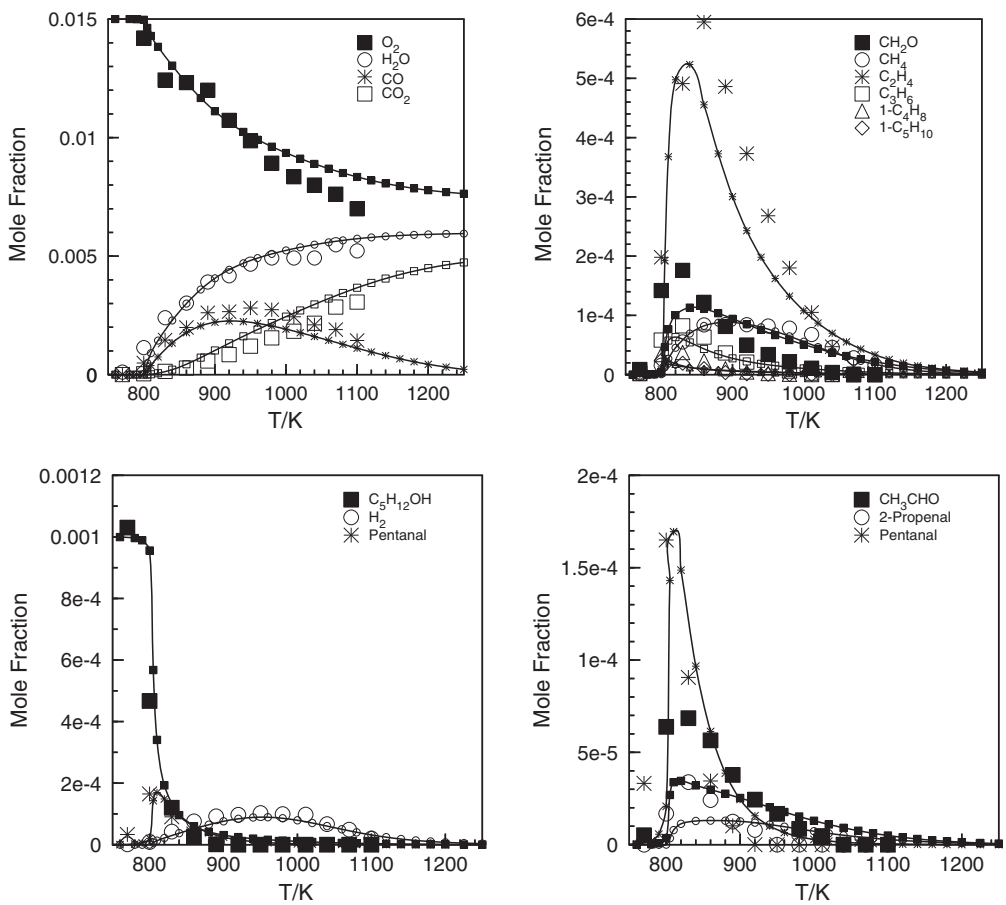

Fig. 2. Experimental (large symbols) and computed (lines and small symbols) concentration profiles obtained from the oxidation of 1-pentanol in a JSR at $\phi = 0.5$, $P = 10$ atm, $\tau = 0.7$ s.

0.5, 1, 2, and 4), the fuel initial concentration was 1000 and 1500 ppm (at $\phi = 4$, results not shown), the temperature ranged from 770 to 1220 K. A good repeatability was observed in the experiments and reasonable good carbon balance of 100 ± 15% was achieved. The concentration profiles of the reactants (1-pentanol and O_2) and products (hydrogen, H_2O, CO, CO_2, CH_2O, CH_4, C_2H_4, C_2H_6, C_2H_2, C_3H_6, CH_3CHO, 2-propenal, propanal, and pentanal) were measured by FTIR and gas chromatography. As can be seen from Figs. 1–4, the consumption of the fuel occurs at >770 K accompanied with a rapid formation of pentanal, formaldehyde, and ethylene. Ethylene and methane are the main intermediate hydrocarbons formed. Their maximum concentration increases as equivalence ratio. Formaldehyde, pentanal, acetaldehyde, and acrolein are the main aldehydes formed during 1-pentanol oxidation. These results are in line with those obtained in similar conditions with 1-butanol [15,18]. Since these chemicals are among the potential pollutants emitted from incomplete fuel combustion, we have compared their maximum intermediate concentrations obtained in a JSR operating at 10 atm, 0.7 s, and $\phi = 1$ [4,18]. In these experiments the initial carbon content was 4000 ppm for 1-butanol and ethanol, and 5000 ppm for 1-pentanol. Therefore, the data were scaled for 4000 ppm of carbon. The data show that all three alcohols produce similar intermediate concentration of formaldehyde (130 ppm for ethanol, 148 ppm for 1-butanol, and 135 for 1-pentanol; uncertainties of 10%) whereas ethanol produces much larger quantities of acetaldehyde (254 ppm for ethanol, 59 ppm for 1-butanol, and 54 for 1-pentanol; uncertainties of 10%). No low-temperature oxidation of 1-pentanol was observed (550–770 K). Therefore, this chemistry was not included in the 1-pentanol kinetic reaction sub-mechanism. Overall, the model represents reasonably well the present data. The intermediate formations of CO and ethylene tend to be underestimated. This is also the case for acetaldehyde under very fuel-lean conditions.

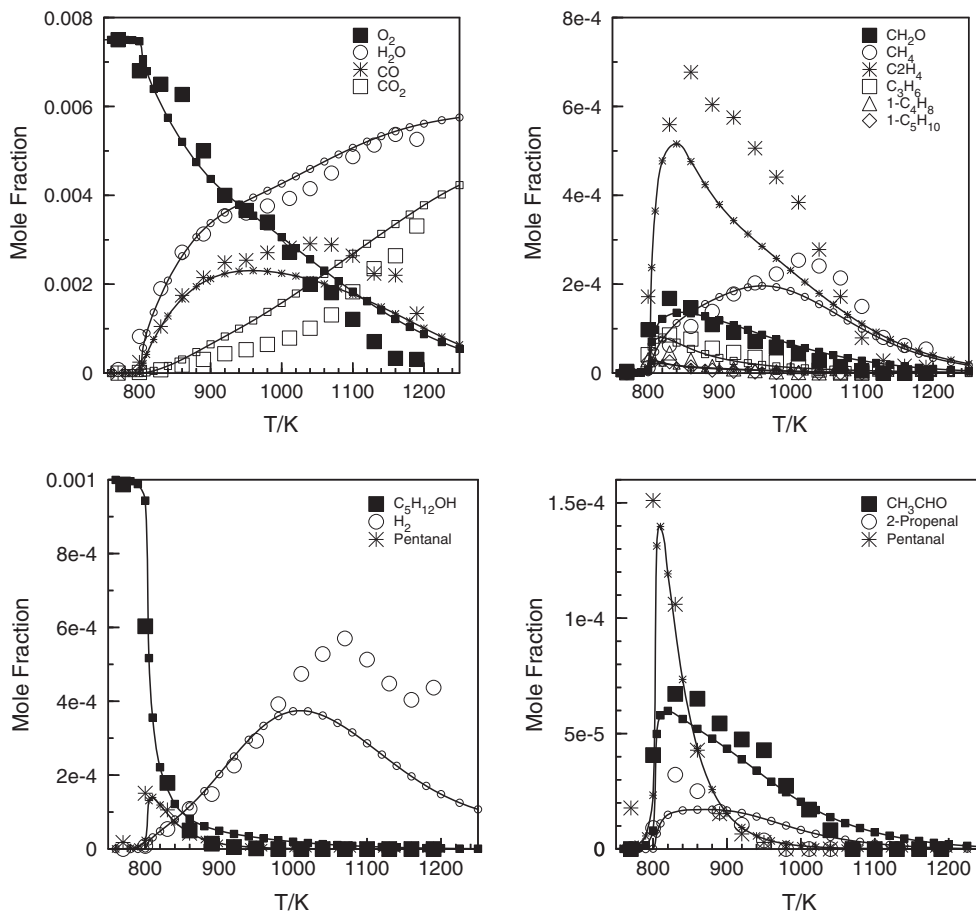

Fig. 3. Experimental (large symbols) and computed (lines and small symbols) concentration profiles obtained from the oxidation of 1-pentanol in a JSR at $\phi = 1$, $P = 10$ atm, $\tau = 0.7$ s.

Reaction paths analyses were performed by computing normalized rates of formation and consumption for every species. According to the proposed model, in stoichiometric conditions, at low fuel conversion (760 K), 1-pentanol reacts via H-atom abstraction by OH (46.4%) and HO_2 (50%). At ca. 65% fuel consumption (810 K), 1-pentanol oxidation occurs via metathesis with OH, mainly yielding a–d$C_5H_{10}OH$ (17.6% each) whereas 14.7% yielded $C_5H_{10}O$ and 8.6% yielded e$C_5H_{10}OH$. The formation of pentanal is mostly due to oxidation of a$C_5H_{10}OH$ by O_2 (98%) whereas its consumption occurs mainly via H-atom abstraction by OH and HO_2 yielding COC_4H_9 that in turn decomposes to form ethylene. Acetaldehyde formation occurs mostly via the decomposition of a$C_5H_{10}OH$. Its consumption occurs via reaction with OH (58%) and HO_2 (24%), yielding CH_3CO that in turn decomposes to yield CO. At higher fuel consumption (98%, 1000 K), the fuel still mainly reacts by metathesis with OH (72%), and to some extent with H (19.8%), and O (5.6%). A schematic representation of 1-pentanol oxidation routes is given in Fig. 5.

4.2. Laminar flame speed and Markstein length

The proposed model was also validated against experimentally measured laminar burning velocities. This experiment allows the study of the combustion of 1-pentanol in a premixed flame environment. The laminar burning velocity was determined over a range of equivalence ratios (0.7–1.4). The error in the flame speed measurements were ±2 cm/s.

Figure 6a compares the experimental and simulated laminar burning velocities at a number of equivalence ratios. The experimental data obtained in this study indicates that the burning velocity increases between $\phi = 0.7$ and $\phi = 1$, which corresponds to a maximum burning velocity of 62 cm/s, and then decreases at higher equivalence ratios. The model well predicts the change in laminar burning velocity as the equivalence ratio is increased from 0.7 to 1.4. The maximum

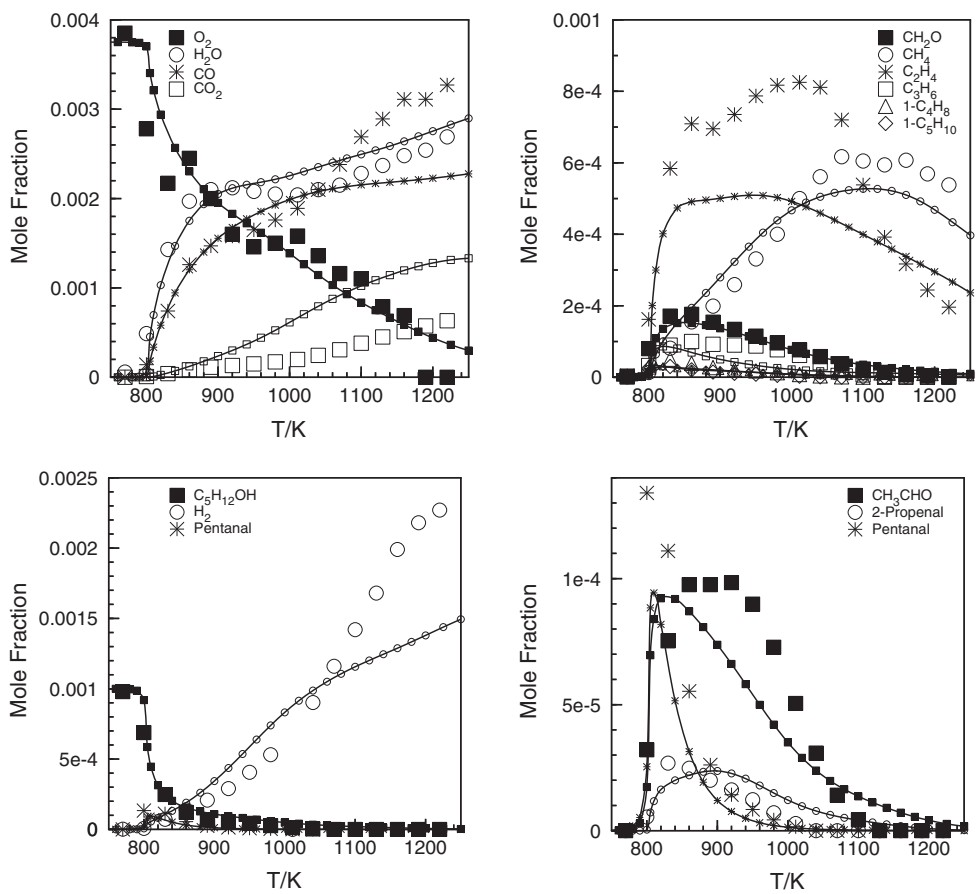

Fig. 4. Experimental (large symbols) and computed (lines and small symbols) concentration profiles obtained from the oxidation of 1-pentanol in a JSR at $\phi = 2$, $P = 10$ atm, $\tau = 0.7$ s.

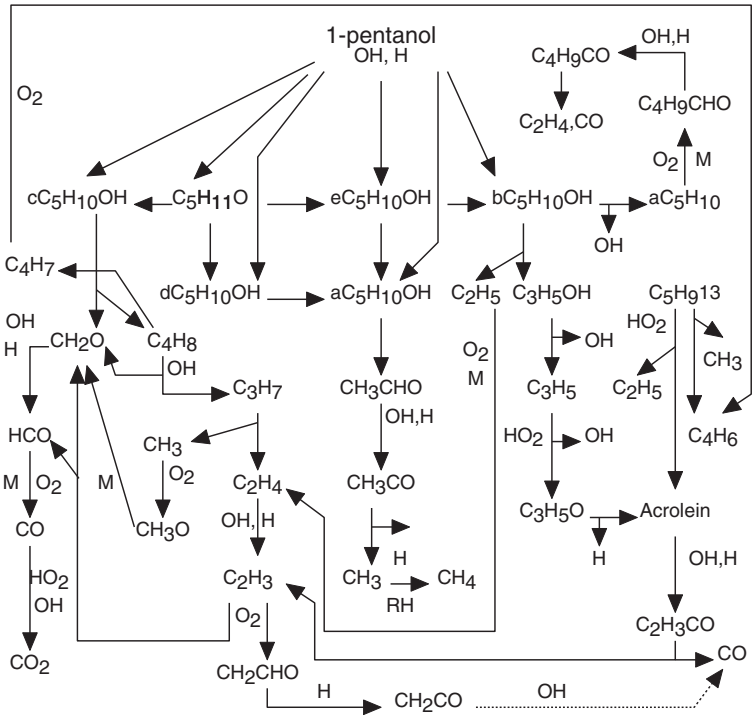

Fig. 5. Reaction paths from the kinetic modeling of 1-pentanol oxidation in a JSR at 10 atm.

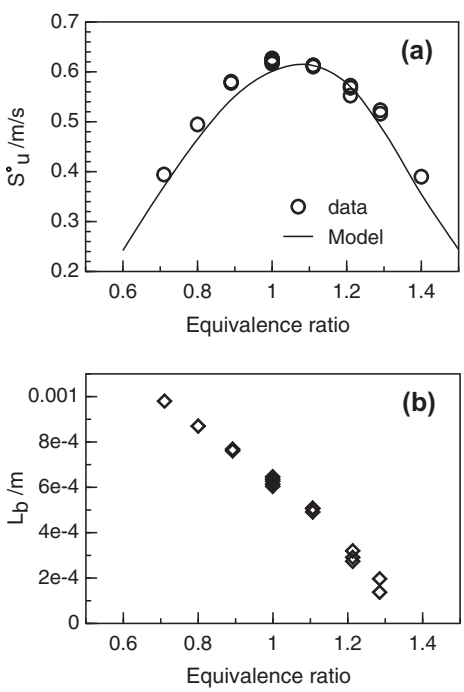

Fig. 6. Laminar burning velocities of 1-pentanol/air mixtures at $T = 423$ K and 1 atm (a) and burnt gases Markstein lengths (b).

burning velocity predicted by the model is 61.4 cm/s at $\phi = 1$, which is three percents less than the experimentally obtained value. Thus, we observe that the proposed model performs well in premixed flame environments.

The burned gas Markstein length, characteristic value of the flame sensibility to stretch, is plotted in Fig. 6b. A decrease is observed when equivalence ratio is increased. The obtained values remain positive indicating that, as observed experimentally, flame fronts are not affected by thermo-diffusive instabilities, whatever the equivalence ratio condition. Moreover, knowing the inherent sensitivity of the Markstein length determination to the image and post-processing qualities, a coherent trend in Markstein length is generally a good indicator of the quality of the laminar burning speed determination.

5. Conclusion

New experimental results were obtained for the oxidation for 1-pentanol in a jet-stirred reactor and a bomb. Concentration profiles of reactants, stable intermediate species, and products were measured in a JSR at 10 atm over a range of equivalence ratios and temperatures. Laminar flame speeds were obtained for 1-pentanol/air

premixed laminar flames at 1 atm. The oxidation of 1-pentanol in the aforementioned experimental configurations was modeled using an extended detailed chemical kinetic mechanism (2099 reactions involving 261 species) derived from a previously proposed scheme for the oxidation of alcohols-based fuels. The proposed mechanism shows good agreement with the present experimental data obtained in a jet-stirred reactor and a bomb. Reaction path analyses were conducted to interpret the results from the JSR experiments.

Appendix A. Supplementary data

This publication includes the following supplemental material: the proposed chemical kinetic reaction mechanism in CHEMKIN format, the associated thermochemical data in CHEMKIN format and the JSR data. Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.003.

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